Change and repair report

1. Integrated the full compose_eos 3.3.0 generator release, matched-EOS provenance, seminar Fortran source, and the completed Cowling spectrum into one relocatable project.
2. Replaced hard-coded absolute-path preparation logic with an explicit CLI and import of the bundled EOS generator API.
3. Added an explicit q-linear adapter builder with exact pressure integration, Euler energy reconstruction, first-law verification, sound-speed proof checks, cubic-Hermite equivalence checks, SHA-256 receipts and PAV-change metrics.
4. Corrected the scientific description: all 234 input q knots are already monotone and the PAV procedure pools zero knots. Pressure re-integration and energy reconstruction, rather than q pooling, are the changes responsible for the nontrivial source/adapter differences.
5. Expanded the self-contained theory and implementation documentation, updated the Cowling-variant source attribution, and added primary references for sharp-interface conversion conditions.
6. Added tests for synthetic EOS construction, thermodynamic identity, causality, no-op PAV and reconstruction of an existing cohort table.
7. Added a read-only 234-row spectrum verifier. No f/p1 integrations were rerun.

The source EOS records remain preserved. The adapter is an additional solver representation, not an overwrite of the EOS generator output.